Number Patterns

Name _____ Date _____ No calculator Version 1

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KEY WORDS pattern *(tauira)* · term *(tūnga)* · position · constant increase · constant decrease · rule *(ture)*

A The next terms ●○○

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 3, 7, 11, 15, ___, ___

b) 5, 11, 17, 23, ___, ___

c) 20, 17, 14, 11, ___, ___

d) 2, 11, 20, 29, ___, ___

e) 50, 43, 36, 29, ___, ___

f) 1, 3.5, 6, 8.5, ___, ___

g) 8, 19, 30, 41, ___, ___

h) 100, 85, 70, 55, ___, ___

B Constant increase or decrease ●○○

2. What is added or subtracted each time?

a) 4, 7, 10, 13, 16

b) 10, 18, 26, 34, 42

c) 30, 25, 20, 15, 10

d) 7, 7.5, 8, 8.5, 9

e) 45, 36, 27, 18, 9

f) 2, 15, 28, 41, 54

C Patterns from diagrams ●●○

3. Squares are made in a row from matchsticks.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of matchsticks					

b) How many matchsticks are added each time?

c) How many matchsticks are there in pattern 10?

d) Describe the rule in words.

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D Rules in words ●●○

Remember: A rule links the position of a term to its value. For 5, 8, 11, 14 the rule is "multiply the position by 3, then add 2", because $3 \times 1 + 2 = 5$ and $3 \times 2 + 2 = 8$.

4. Describe the rule that links the position to the term.

a)

Position	1	2	3	4
Term	2	4	6	8

Rule: _____

b)

Position	1	2	3	4
Term	5	8	11	14

Rule: _____

c)

Position	1	2	3	4
Term	6	11	16	21

Rule: _____

d)

Position	1	2	3	4
Term	3	7	11	15

Rule: _____

E Rules as equations ●●●

Remember: Use n for the position and T for the term. "Multiply the position by 3, then add 2" is written $T = 3n + 2$.

5. Write the rule for each pattern as an equation.

a) 5, 8, 11, 14, 17

b) 7, 9, 11, 13, 15

c) 4, 8, 12, 16, 20

d) 1, 6, 11, 16, 21

e) 10, 16, 22, 28, 34

f) 9, 19, 29, 39, 49

F Using a rule ●●●

Remember: In the rule $T = 3n + 2$, put the position in place of n. For the 4th term: $3 \times 4 + 2 = 14$.

6. Write the first four terms of the pattern with each rule.

a) $T = 3n + 2$

b) $T = 5n - 1$

c) $T = 2n + 7$

d) $T = 4n$

7. Find the term asked for.

a) $T = 3n + 2$, the 10th term

b) $T = 5n - 1$, the 20th term

c) $T = 4n + 7$, the 50th term

d) $T = 2n + 5$, the 100th term

★ Extension

8. A pattern has the rule $T = 3n + 2$. Which position has the term 32?

9. The pattern 40, 37, 34, 31, … keeps going down by 3. Is 10 a term of the pattern? Show how you know.

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Number Patterns

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS pattern *(tauira)* · term *(tūnga)* · position · constant increase · constant decrease · rule *(ture)*

A The next terms ●○○

Remember: In a linear pattern the same amount is added or subtracted each time.

1. Write the next two terms of each pattern.

a) 2, 7, 12, 17, ___, ___

b) 4, 11, 18, 25, ___, ___

c) 25, 21, 17, 13, ___, ___

d) 3, 11, 19, 27, ___, ___

e) 60, 51, 42, 33, ___, ___

f) 2, 3.5, 5, 6.5, ___, ___

g) 7, 19, 31, 43, ___, ___

h) 90, 75, 60, 45, ___, ___

B Constant increase or decrease ●○○

2. What is added or subtracted each time?

a) 5, 9, 13, 17, 21

b) 12, 21, 30, 39, 48

c) 40, 34, 28, 22, 16

d) 3, 3.5, 4, 4.5, 5

e) 54, 46, 38, 30, 22

f) 3, 17, 31, 45, 59

C Patterns from diagrams ●●○

3. Dots are drawn in the shape of a letter L.



Pattern 1 **Pattern 2** **Pattern 3**

a) Complete the table.

Pattern	1	2	3	4	5
Number of dots					

b) How many dots are added each time?

c) How many dots are there in pattern 10?

d) Describe the rule in words.

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D Rules in words ●●○

Remember: A rule links the position of a term to its value. For 5, 8, 11, 14 the rule is "multiply the position by 3, then add 2", because $3 \times 1 + 2 = 5$ and $3 \times 2 + 2 = 8$.

4. Describe the rule that links the position to the term.

a)

Position	1	2	3	4
Term	3	6	9	12

Rule: _____

b)

Position	1	2	3	4
Term	7	11	15	19

Rule: _____

c)

Position	1	2	3	4
Term	7	12	17	22

Rule: _____

d)

Position	1	2	3	4
Term	5	11	17	23

Rule: _____

E Rules as equations ●●●

Remember: Use n for the position and T for the term. "Multiply the position by 3, then add 2" is written $T = 3n + 2$.

5. Write the rule for each pattern as an equation.

a) 6, 10, 14, 18, 22

b) 5, 7, 9, 11, 13

c) 3, 6, 9, 12, 15

d) 2, 7, 12, 17, 22

e) 9, 16, 23, 30, 37

f) 8, 18, 28, 38, 48

F Using a rule ●●●

Remember: In the rule $T = 3n + 2$, put the position in place of n. For the 4th term: $3 \times 4 + 2 = 14$.

6. Write the first four terms of the pattern with each rule.

a) $T = 4n + 1$

b) $T = 3n - 2$

c) $T = 2n + 9$

d) $T = 6n$

7. Find the term asked for.

a) $T = 4n + 1$, the 10th term

b) $T = 3n - 2$, the 20th term

c) $T = 6n + 5$, the 50th term

d) $T = 2n + 3$, the 100th term

★ Extension

8. A pattern has the rule $T = 4n + 1$. Which position has the term 41?

9. The pattern 40, 37, 34, 31, … keeps going down by 3. Is 20 a term of the pattern? Show how you know.

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